

## 47.7

STUDIES ON SUBFRAGMENTS OF NORMAL ATRIAL AND THYROTOXIC VENTRICULAR MYOSIN. Surath K. Banerjee\*, Sandra Livernais\* and George Kaldor. Clinical Laboratory Service, VA Medical Center, Allen Park, MI 48101 and Department of Pathology, Wayne State University, School of Medicine, Detroit, MI 48202.

Expression of atrial myosin, unlike ventricular myosin, is not modulated by thyroid hormone. In order to explore the mechanism for the differential effects of thyroid hormone we chose to study subfragment of normal atrial myosin and compare it to that of V<sub>1</sub>-isomyosin which is expressed in thyrotoxic ventricle. Subfragment-1 of atrial and V<sub>1</sub> ventricular myosin were prepared and purified in a DEAE-cellulose column chromatography. Although atrial muscle is known to contain equimolar amounts of A<sub>1</sub> and A<sub>3</sub> isomyosin, only a single component of atrial subfragment was obtained. Pyrophosphate-polyacrylamide gel electrophoretic patterns and column chromatographic profile of this atrial subfragment differ from those of V<sub>1</sub>-isomyosin. On the other hand, Ca<sup>2+</sup> and actin activated ATPase activities of these subfragments are identical. Comparison of peptide mapping after limited proteolysis of the heavy and the light chain, reveals identical patterns for the heavy chain peptides of these subfragments but the light chain patterns differ. These results suggest that the heavy chain of atrial subfragment may have identical structure and function as that of V<sub>1</sub>-isomyosin. The differences between these subfragments are due to the differences in the nature of light chains associated with them. Supported by VA Medical Research Fund and MHA.

## 47.9

IN VIVO <sup>31</sup>P-NMR IN HUMAN MUSCLE: GATING OF METABOLIC PHOSPHATES AND pH OVER THE CONTRACTION-RELAXATION CYCLE OF STEADY STATE WORK. P. A. Molé, R. L. Coulson, and J. R. Caton. University of California, Davis, CA 95616 and Southern Illinois University, Carbondale, IL 62901

Using <sup>31</sup>P-NMR in vivo, concentrations of metabolic phosphates and pH were measured at 15 discrete instances over the contraction-relaxation cycle of human flexor digitorum superficialis muscle performing rhythmic steady state work. A handgrip ergometer adjusted to a work load of 1.5 J was used, and contractions were of 1 sec duration and repeated every 2 sec. After exercise had been maintained for 5 mins, <sup>31</sup>P-NMR resonance signals representing 16 scans were averaged at each of the 15 time intervals using an Oxford TMR-32 spectrometer and a gating trigger which matched average <sup>31</sup>P resonances to specific times during the contraction-relaxation cycle. Preliminary observations indicate there were significant variations in [ATP], [PCr], [Pi], and [H<sup>+</sup>], with work within the contraction cycle. Two important patterns of variation were noted; one proportional to work done for PCr and Pi, and the other proportional to the rate of change in work for PCr, Pi, ATP, and H<sup>+</sup>. Thus, enzymatic control of energetic reactions of muscle contraction and relaxation appear to contain both load-sensitive and rate-sensitive control elements.

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## 47.8

IN VIVO <sup>31</sup>P-NMR MEASUREMENT OF METABOLIC PHOSPHATES AND pH IN HUMAN MUSCLE DURING WORK WITH CIRCULATION OCCLUDED. R. L. Coulson, P. A. Molé, and J. R. Caton. Southern Illinois University, Carbondale, IL 62901 and University of California, Davis, CA 95616.

With circulation to the right forearm occluded by a tourniquet, [PCr], [Pi], [ATP], and pH were measured following successive contractions of the flexor digitorum superficialis muscle. Human subjects performed work of 1.5 J on a hand-grip ergometer where contractions were of 1 s duration and repeated every 2 min until voluntary contractions could no longer be elicited. Eighteen contractions were performed in 36 min. Sixteen NMR scans, using an Oxford TMR-32 spectrometer were accumulated immediately after each contraction and again 1 min later. There was no change in any of the quantities measured during the minutes in which no contraction occurred. There was no change in [ATP] at any time or with any contraction. During the first 6 contractions at this work load, [PCr], [Pi], and pH were maintained. Thereafter, [PCr] declined and [Pi] rose by similar amounts with each contraction. pH did not fall suggesting that this amount of work does not tax the buffer capacity of the muscle groups involved.

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## 47.10

ELEVATED NUCLEAR MAGNETIC RESONANCE RELAXATION TIMES IN CARDIOMYOPATHY. L.K. Misra\*, T.F. Egan\*, D.W. Bearden\*, D. Roy\*, S.H. Kim\*, and C.F. Hazlewood. Departments of Physiology and Pathology, Baylor College of Medicine, Houston, TX 77030, and Physics Department, Rice University, Houston, Tx.

Cardiomyopathy in the Bio 14.6 strain of Syrian hamsters progresses in four well-characterized phases. Our objective was to test the hypothesis that various clinical and histologic phases of cardiomyopathy cause specific alterations in the proton nuclear magnetic relaxation (NMR) times of cardiac tissue. Cardiomyopathic and control hamsters were sacrificed at 190 days of age and left ventricles were excised. Sections were stained with hematoxylin-eosin and trichrome for light microscopy. The spin lattice relaxation times, T<sub>1</sub>, were measured at 30 MHz and 80 MHz. The spin-spin relaxation times, T<sub>2</sub>, were measured at 80 MHz.

Multifocal scanning of varying cellularity and collagen accumulation were observed in the left ventricular free wall and septum of cardiomyopathic hamsters. Admixed were also foci of dystrophic calcifications. A significant elevation in T<sub>1</sub> values was seen in the ventricles of the cardiomyopathic hamsters. The T<sub>1</sub> values of cardiomyopathic and control hamsters were 990±29 ms and 944±11 ms, respectively, at 80 MHz, and 667±16 ms and 629±17 ms, respectively, at 30 MHz. The resolution power of T<sub>1</sub> at 30 MHz (5.58) was higher than that at 80 MHz (4.71). In T<sub>2</sub>, a significant difference in the long-fracture was found between the control (58±9 ms) and cardiomyopathic (98±16 ms) hamsters. These results show that changes in the NMR parameters are useful in detection of pathologic events of heart. (Supported by R01-AM35479)

## SMOOTH MUSCLE

## 48.1

ESTROGENS AND ENDOTHELIUM-DEPENDENT RESPONSES IN THE FEMORAL ARTERY OF THE RABBIT. Veronique Gisclard\* and Paul M. Vanhoutte. Dept. Physiol., Mayo Clinic, Rochester, MN 55905

Experiments were designed to determine the chronic effect of estrogens on vascular responsiveness to alpha-adrenergic activation and endothelium-dependent relaxations to acetylcholine. Female rabbits were oophorectomized and given either 17-β-estradiol (100 µg for 4 days) or solvent. One day after the last injection, the animals were sacrificed, and the femoral arteries excised. Rings with and without endothelium were suspended for isometric tension recording in organ chambers filled with physiological salt solution. In the control group, removal of the endothelium shifted the concentration-response curve to norepinephrine to the left. Treatment with 17-β-estradiol caused a shift to the right of the concentration response curve to the catecholamine, and abolished the effect of endothelium-removal. Inhibition of neuronal and extraneuronal uptake and of beta-adrenergic receptors, reduced responses to lower concentrations of norepinephrine in the treated, but not in the control arteries with endothelium. Treatment with 17-β-estradiol potentiated the relaxations evoked by acetylcholine in femoral arteries with endothelium. These experiments suggest that chronic treatment with estrogens enhance endothelium-dependent inhibitory responses to alpha-adrenergic and cholinergic activation in arteries of the rabbit. (Supported in part by NIH grant HL 31183.)

## 48.2

VASCULAR BETA-ADRENORECEPTOR ACTIVITY IS ALTERED IN HYPERTHYROID AND HYPOTHYROID RATS. D. B. STRATTON. Dept of Biology, Drake University, Des Moines, IA 50311

Young male rats were treated daily for two weeks either with 200 µg injections of L-thyroxine (TRX) or 0.1% propylthiouracil (PTU) in their drinking water. Rings of thoracic aorta (2 mm) were mounted in muscle baths and isometrically contracted with 10<sup>-12</sup> to 10<sup>-4</sup> M norepinephrine (NE). NE concentrations above 5 x 10<sup>-8</sup> M generated significantly less ring tension from TRX rats than from euthyroid control rats. Rings from PTU rats generated more tension than rings from controls at the highest concentrations of NE but not significantly so. All rings were subsequently precontracted to 70% maximum tension with NE and then sequentially relaxed with 10<sup>-10</sup> to 10<sup>-7</sup> M sodium nitroprusside, a nonspecific vasodilator, or 10<sup>-8</sup> to 10<sup>-5</sup> M isoproterenol, a beta-adrenoreceptor agonist. No difference was observed in the relaxation curves to sodium nitroprusside but significantly greater ring relaxation was observed from TRX rats and significantly less relaxation was seen from PTU rats when compared to controls. These results suggest that thyroxine treatment increases vascular beta-adrenoreceptor activity while propylthiouracil treatment decreases beta-adrenoreceptor activity in rat thoracic aorta.